

Your grade: 100%

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Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1. For a scalar variable x , what is $\int_{-\infty}^{\infty} x^2 \delta(x - 1) dx$?

1 / 1 point

1

✔ Correct
Yes, that is correct.

2. For vector variable x having two elements, what is:

1 / 1 point

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x^T x) \delta(x - x_0) dx_1 dx_2,$$

if:

$$x_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

5

✔ Correct
Yes, that is correct.

3. Suppose that $g(x) = x^2$ and $\hat{f}(x) = \frac{1}{3} \sum_{i=1}^3 \delta(x - x^{(i)})$, where $x^{(1)} = 0.1$, $x^{(2)} = 0.5$, and $x^{(3)} = 1$. What is the value of $\hat{\mu} = \int_{-\infty}^{\infty} g(x) \hat{f}(x) dx$?
(Enter your answer with two digits to the right of the decimal place.)

1 / 1 point

0.42

✔ Correct
Yes, that is correct.